

Assignment - 01

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1. Define Machine Learning, explain its relationship with Artificial Intelligence and deep learning with suitable examples.

→ A computer program is said to learn from experience E with respect to some class of tasks T and performance at tasks in T , as measured by P , improves with experience E .

There are three important components:

1. Task (T)

The task represents what the machine learning system has to perform.

Ex:

- Prediction
- Classification
- Acting

2. Experience (E)

Experience means the data or previous examples from which the machine learns

3. Performance Measure (P)

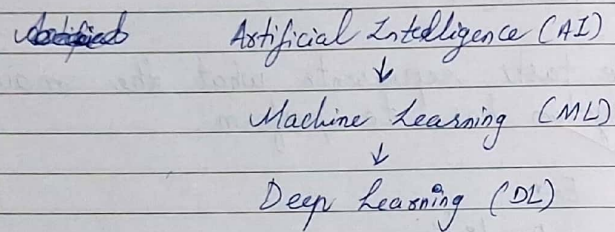
Performance measure tells us how well the machine learning system is performing

Ex:-

- Accuracy
- Error
- Mean Squared Error
- Precision
- Recall

* Relationship between AI, ML and deep learning.

The relationship can be represented as :



A. Artificial Intelligence (AI)

AI is the broad field of creating machines or computer systems that can perform tasks that normally require human intelligence.

Ex:-

- Voice assistants.
- Game-playing systems.
- Expert systems.

B. Machine Learning (ML) :-

ML is a subset of AI, It allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed with every rule.

Ex:-

A bank can use ML to analyze previous transactions and detect whether a new transaction is potentially fraudulent.

Common types of ML :-

1. Supervised Learning - Learns from labeled data.
2. Unsupervised Learning - finds patterns in unlabeled data.
3. Reinforcement Learning - Learns through rewards and penalties.

C. Deep Learning (DL)

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers to learn complex patterns from large amounts of data.

It is especially useful for:-

- Image recognition
- Speech recognition
- Natural Language processing
- Computer Vision

Ex:- A face-recognition System can use a deep neural network to learn features such as edges, shapes, eyes, nose and overall facial patterns.

3. Diff between AI, ML, and DL

Feature	AI	ML	DL
Meaning	Makes machines intelligent	Machines learn from data	uses deep neural networks to learn
Scope	Broadest Concept	Subset of AI	Subset of ML
Data requirement	May or may not require large data	Usually requires data	Usually requires large amounts of data
Human involvement	Rules may be manually designed	Model learns patterns	Automatically learns complex features

2. Explain the types of Machine learning: Supervised, Unsupervised, Semi-supervised and Reinforcement learning with examples.

→ Machine learning can be broadly classified into four main types:

1. Supervised learning.
2. Unsupervised learning.
3. Semi-supervised learning.
4. Reinforcement Learning.

The main difference is how the machine learns from data:-

1. Supervised Learning:-

Supervised learning is a type of machine learning in which the model is trained using labeled data. The input data and the correct output are already provided.

The model learns the relationship between input and output and uses it to predict the output for new data.

Ex:-

- Email spam detection - Spam/Not Spam.
- House price prediction
- Disease classification

2. Unsupervised Learning:

Unsupervised learning is a type of machine learning in which the model is trained using unlabeled data. There is no predefined correct output.

The model tries to discover hidden patterns, groups or relationships in the data.

Ex:-

- Customer segmentation.
- Grouping similar documents.
- Finding unusual patterns in data.
- Market basket analysis.

Ex:-

A shopping company has information about thousands of customers but does not have predefined customer groups. A unsupervised learning algorithm can group customers based on their purchasing behavior, such as:

- Group 1 → frequent buyers.
- Group 2 → occasional buyers.
- Group 3 → Low-frequency buyers.

3. Semi-Supervised Learning:-

def:-

semi-supervised learning uses a combination of a small amount of labeled data and a large amount of unlabeled data for training.

It is useful when obtaining labeled data is expensive or time-consuming, but large amounts of unlabeled data are easily available.

Ex:-

Suppose a company has 10,000 images, but only 500 images are labeled as cats or dogs. The remaining 9,500 images are unlabeled.

Semi-supervised learning can use both the labeled and unlabeled images to build a better model.

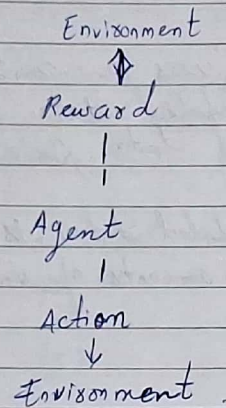
4. Reinforcement Learning:-

def:-

Reinforcement learning is a type of machine learning in which an agent learns by interacting with an environment and receives rewards or penalties for its actions.

The goal is to learn a ~~strategy~~ strategy that maximizes the total reward over time.

Basic Components :-



Ex:-

In a game, an AI agent tries different moves. If it makes a good move, it receives a positive reward. If it makes a bad move, it receives a penalty. After many attempts, it learns which actions are better.

Applications:-

- Game playing
- Robot Control
- Autonomous vehicle
- Resource Management

37 What is Hypothesis space? Explain its concept with a suitable ~~exple~~ example.

→ In Machine Learning, Hypothesis space is the set of all possible hypotheses (models or rules) that a learning algorithm can choose from to solve a particular problem.

A hypothesis is a possible function or rule that maps input data to an output.

The learning algorithm searches this space to find the best hypothesis that fits the training data.

Concept of Hypothesis Space:-

✓ Suppose we want to predict whether a student will pass or fail based on the number of hours they study.

We can have different possible hypotheses:

- h_1 : If study hours $> 2 \rightarrow$ pass, otherwise Fail.
- h_2 : If study hours $> 4 \rightarrow$ pass, otherwise Fail.
- h_3 : If study hours $> 6 \rightarrow$ pass, otherwise Fail.
- h_4 : Always $\rightarrow$ pass
- h_5 : Always $\rightarrow$ Fail.

- All these possible rules together form the hypothesis space.

Hypothesis space

The ML algorithm evaluates these hypotheses using the training data and selects the one that performs best.

Ex: Consider the following training data:

Study hours	Result
1	Fail
2	Fail
3	pass
4	pass
5	pass.

possible hypotheses could be:

- h_1 : pass if hours ≥ 2
- h_2 : pass if hours ≥ 3
- h_3 : pass if hours ≥ 4
- h_4 : Always pass.
- h_5 : Always Fail

From the training data, h_2 (pass if study hours ≥ 3) correctly classifies all the given examples.

Therefore, h_2 is a suitable hypothesis selected from the hypothesis space.

* Importance of Hypothesis Space :-

1. It defines the possible solutions that a learning algorithm can consider.
2. It helps the algorithm search for the best model.
3. A larger hypothesis space provides more possible models but may make learning more difficult.
4. A very small hypothesis space may not contain a suitable model.
5. The choice of hypothesis space affects the accuracy and generalization of the ML model.

Q. What is Inductive Bias?

→ Inductive Bias is the set of assumptions or preferences that a machine learning algorithm uses to make predictions for new, unseen data.

Since a model usually cannot learn every possible pattern from limited training data, it uses some assumptions to select a suitable hypothesis.

- Why is Inductive Bias Needed?

Suppose a model is given only a few training examples. There can be many different hypotheses that correctly classify those examples.

✓ The algorithm needs some assumptions to decide which hypothesis is more likely to work on new data.

Therefore, inductive bias helps the model generalize from training data to unseen data.

Ex:

Suppose we have training data showing the following:

Hours Studied	Result
1	Fail
2	Fail
3	Fail
4	Pass
5	Pass

Now, many rules could fit this data.

One possible rule is:

If study hours ≥ 3 , then pass; otherwise Fail

An algorithm may have the bias that simpler rules are preferred over complicated rules.

✓ Therefore, it selects the simple rule rather than creating a complicated rule that exactly memorizes the training example.

This preference for simpler solutions is an example of inductive bias.

5. Explain the different evaluation metrics for regression and classification problems.

→ Evaluation metrics ~~for~~ are measures used to determine how well a Machine Learning model performs on unseen/ test data.

The evaluation metrics are different for regression and classification problems.

A. Evaluation Metrics for regression:-

Regression predicts a continuous numerical value, such as house price, temperature, salary etc.

The most commonly used regression metrics are:

1. Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where :

- y_i = actual value.
- $\hat{y}_i$ = predicted value.
- n = number of observations.

Lower MAE = Better model

Ex:-

If MAE = ₹ 5,000, the model's prediction is wrong by an average of about ₹ 5,000.

2. Mean Squared Error (MSE)

MSE calculates the average of the squared differences between actual and predicted values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

It gives more importance to large errors because the errors are squared.

Lower MSE = Better Model

3. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

It is useful because RMSE is expressed in the same units as the target variable

Lower RMSE = Better Model.

4. R^2 score (Coefficient of Determination)

R^2 measures how well the model explains the variation in the dependent variable.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

where:

- SS_{res} = Sum of squared residuals.
- SS_{tot} = total sum of squares.

an R^2 value closer to 1 generally indicates a better fit.

Ex:

If $R^2 = 0.90$ the model explains about 90% of the variation in the target variable.

B Evaluation Metrics for Classification

classification predicts a category or class, such as:

- Spam / Not spam.
- pass / Fail.

- Disease / No Disease.

classification evaluation is based on a confusion matrix.

Confusion Matrix:-

A confusion matrix contains four imp terms:

	Predicted positive	Predicted Negative
Actual positive	TP (True positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

1. Accuracy.

✓ Accuracy measures the ~~program~~ proportion of correct predictions among all predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Higher accuracy = Better model

2. Precision.

Precision measures how many of the instances predicted as positive are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

3. Recall (sensitivity)

Recall measures how many of the actual positive ~~cases~~ cases are correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall means fewer false negatives.

4. F1-Score

F1-score is the harmonic mean of precision and Recall.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It is useful when we want a balance between precision and recall.

Higher F1-score = Better model

6. What is cross-validation?

→ Cross-validation is a model evaluation technique in machine learning used to determine how well a model will perform on unseen data.

In cross-validation, the available dataset is divided into multiple parts called folds. The model is trained and tested multiple times using different combinations of these folds.

Cross-validation is used to:

1. Evaluate the generalization ability of a model
2. Reduce the dependency on a single train-test split.
3. Detect overfitting.
4. Compare different Machine learning models.
5. Select suitable hyperparameters.

7. Explain the Decision Tree Algorithm. Describe its structure and working with a suitable example.

→ A Decision Tree is a supervised Machine learning algorithm used for Classification and regression problems. It makes predictions by repeatedly dividing the data based on certain conditions or features.

It has a tree-like structure, where each internal node represents a decision, each branch represents an outcome of that decision and each leaf node represents the final predictions.

Structure of a Decision Tree :-

A decision tree consists of the following components:

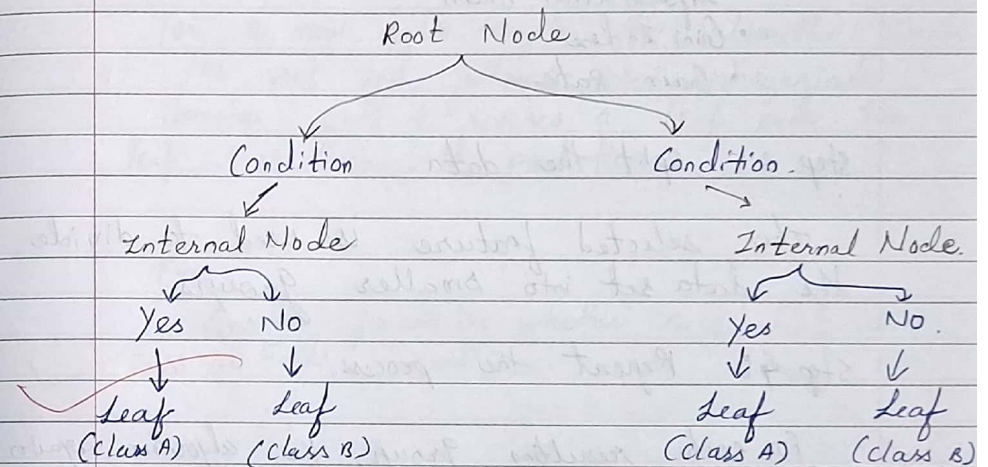
1. Root Node - The topmost node of the tree. It represents the first and most important decision.

2. Internal / Decision Node -

Represents a condition or test on a feature.

3. Branch / Edge - Represents the outcome of a decision.

4. Leaf Node - Represents the final class or predicted value.



Working of Decision Tree :-

The basic working steps are:

Step 1: Start with the complete dataset.

The algorithm starts with all training data at the root node.

Step 2: select the best feature.

The algorithm evaluates different features and selects the one that provides the best split of the data.

Common criteria include :-

- Information Gain
- Gini Index
- Gain Ratio.

Step 3: Split the data.

The selected feature is used to divide the data set into smaller groups.

Step 4:- Repeat the process.

For each resulting group, the algorithm again selects the best feature and creates another decision node.

Step 5: stop splitting

splitting stops when a suitable stopping condition is reached, such as:

- All or most samples belong to the same class.

- No useful feature remains

- Maximum tree depth is reached.

- Minimum number of samples is reached

Step 6: Make prediction

For a new data sample, the algorithm starts at the root and follows the appropriate branches until it reaches a leaf node. The leaf gives the final prediction.

Ex:-

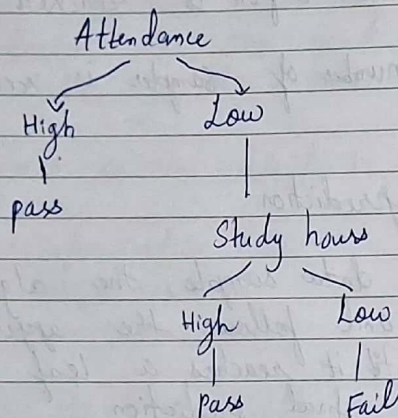
Consider predicting whether a student will pass or Fail based on:

- Attendance
- Study Hours.

Training data:

Attendance	Study Hours	Result
High	High	pass
High	Low	pass
Low	High	pass
Low	Low	Fail.

A possible Decision Tree is :

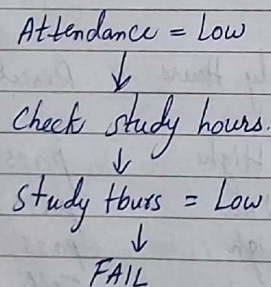


Prediction:-

Suppose a new student has :

- Attendance = Low
- Study hours = Low.

The tree works as follows :



Therefore, the model predicts Fail.

8. What is Overfitting? Explain its causes, effects and methods to prevent overfitting.

→ Overfitting is a situation in Machine Learning where a model learns the training data too closely, including its noise and unnecessary details, instead of learning the general patterns.

As a result, the model performs, very well on training data but poorly on new or unseen data

Ex:-

Suppose we train a Decision Tree to classify whether students will pass or fail.

If the tree becomes extremely deep, it may create rules for almost every individual training student.

Training Data



Very Complex Model.



Training Accuracy = 99%



New / Test Data.



Test Accuracy = 65%

The model has learned the training data too specifically. This is overfitting.

Causes of overfitting:-

1. Complex Model:-

A model with too many parameters or a very deep decision tree can learn unnecessary details.

2. Small Training Dataset:-

If the training dataset is too small the model may not learn the general pattern properly.

3. Noisy Data:-

Noise, errors or incorrect observations in the training data can be learned as if they were meaningful patterns.

4. Too many Features:-

Using a large number of irrelevant or unnecessary features can make the model unnecessarily complex.

5. Excessive training:-

In some models, training for too long can cause the model to start learning noise in the training data.

Effects of overfitting:-

1. High training accuracy but low testing accuracy
2. poor performance on unseen data.
3. poor generalization ability.
4. The model becomes unnecessarily complex.
5. predictions become sensitive to small changes in input data

Methods to prevent overfitting:-

1. Using more training data
2. Cross Validation.
3. Regularization.

- L1 regularization (Lasso)
- L2 regularization (Ridge)

4. By Reducing Model complexity.

9. What is Instance - Based. Learning?

→ Instance Based Learning (IBL) is type of machine learning in which the model stores the training examples (instances) and makes predictions for new data by comparing it with similar stored examples.

It is also called lazy learning, because the model does not build a general model during the training phase. Instead, most of the computation is performed when a new instance needs to be classified or predicted.

How instance based learning works.

The basic steps are:-

1. Store the training data in memory.
2. When a new data point arrives, calculate its similarity or distance from stored examples.
3. Find the most similar instances.
4. Use those instances to make the prediction.

Training Data

↓
Store Instances

↓
New Data Point

↓
Calculate Similarity / Distance

↓
Find Similar Instances

↓
Make prediction.

10. What is feature reduction. Explain its need, advantages and commonly used techniques.

→ Feature Reduction is the process of reducing the number of input features (variables) in a dataset while retaining the most important information needed for Machine Learning.

Need for feature Reduction:-

Feature reduction is needed for the following reasons:-

1. Reduce Computational cost

Fewer features require less memory and computational power.

2. Remove Irrelevant features.

Some features may have little or no useful information for prediction.

Removing them can improve model performance.

3. Reduce Overfitting:

A large number of features can make a model unnecessarily complex and increase the possibility of overfitting.

4. Improve Model performance.

Removing noisy and redundant features can sometimes improve the accuracy of the model.

5. Easier Visualization.

High-dimensional data is difficult to visualize. Reducing it to 2 or 3 dimensions make visualization easier.

6. Faster Training.

Models can be trained faster when the number of input features is reduced.

Advantages of Feature Reduction -

1. Reduces Computational complexity.

2. Reduces memory requirements.

3. Decreases training time.

4. Helps prevent overfitting.

5. Removes irrelevant and redundant information.

6. Makes models easier to understand.

7. Allows visualization of high-dimensional data.

Commonly Used Techniques:-

1. Feature selection

Feature selection selects the most relevant features from the original feature set without changing their original representation.

* Common methods:-

- Filter method
- Wrapper method
- Embedded method.

2. Feature Extraction.

Feature extraction creates new features by transforming or combining the original features.

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- Autoencoders.

11. A student performance prediction system is given. suggest suitable features, Machine Learning algorithm, evaluation method and explain how overfitting can be avoided.

→ A student performance prediction system is a Machine learning system that predicts a student's future academic performance, such as Pass/Fail, grade, or expected marks, using relevant student related data.

1. Suitable Features:-

The following features can be used as input:

Feature	Description
Attendance	percentage of classes attended.
Study hours	Average hours spent studying
previous Marks	Marks obtained in previous exams.
Assignment Marks	performance in assignments.
Internal Assessment	Internal examination / test marks.
practical / Lab performance	performance in practical work
Number of Backlogs	previous failed subjects, if applicable.
class participation	participation in academic activities.

The target variable can be:

- pass / Fail - classification problem
- Expected Marks - Regression Problem.
- Grade (A, B, C) etc → classification problem.

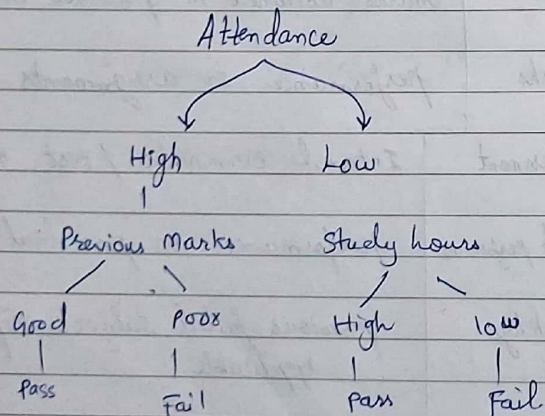
2. Suitable Machine Learning Algorithm.

for pass / fail prediction.

decision tree or random forest can be used.

A Decision tree is easy to understand and can show how factors such as attendance, study hours and previous marks contribute to the prediction.

Ex:-



for predicting numerical marks:-

If the objective is to predict actual marks, linear Regression or another regression algorithm can be used.

3. Evaluation Method:-

The dataset should be divided into training and testing data, for ex:

Student Dataset



Training Data



Train Model → Evaluate Model.

for more reliable evaluation, k-Fold cross-validation can be used

If predicting Pass / Fail

Use classification metrics such as:

- Accuracy
- precision.
- Recall
- F1-score
- Confusion Matrix

For ex, if the model correctly predicts 90 out of 100 students

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$$\text{Accuracy} = \frac{90}{100} = 90\%$$

If predicting marks

use regression metrics such as:

- MAE
- MSE
- RMSE
- R^2 Score

4. How to Avoid Overfitting.

Overfitting occurs when the model performs very well on training students but performs poorly on new students.

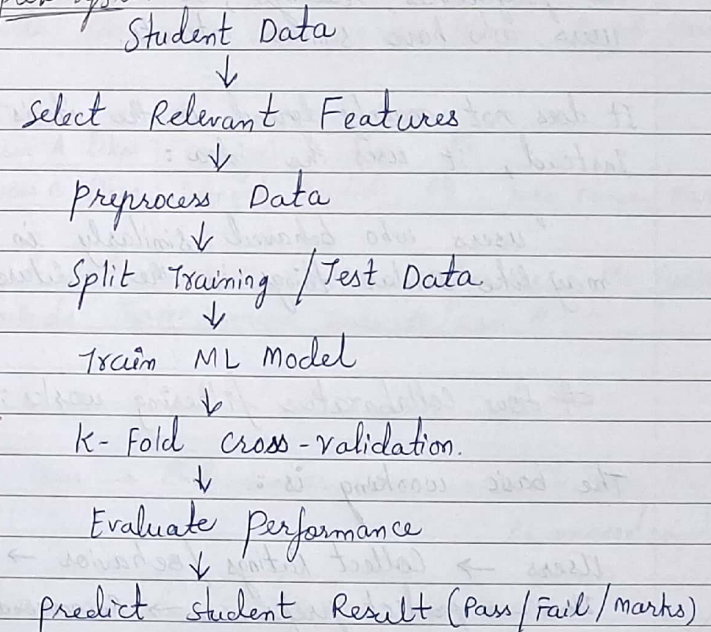
It can be avoided by:

1. Cross validation
2. Use of more training data
3. Feature selection.
4. Limit Model complexity
5. pruning.

6. Regularization.

7. Use a separate Test set.

Complete system:-



12. Explain Collaborative filtering-based recommendation systems with a suitable real-world example.

→ Collaborative filtering (CF) is a recommendation technique that suggests items to a user based on the preferences, ratings, or behaviors of other users who have similar interests.

It does not mainly depend on the item's features. Instead, it uses the idea:

"users who behaved similarly in the past may like similar things in the future."

How Collaborative filtering works:

The basic working is:

Users → Collect Ratings / Behavior → Find Similar Users
Items → predict preference → Recommend Items.

for ex; Suppose a movie platform has information about which movies users watched and liked.

If User A and User B have liked many of the same movies, the system considers them similar.

If user B likes a movie that user A has not watched, the system may recommend that movie to user A.

Types of Collaborative Filtering.

There are mainly two types:

A. User Based Collaborative filtering.

It finds users who are similar to the target user.

Ex:-

- User A likes: Dangal, 3 Idiots, PK
- User B likes: Dangal, 3 Idiots, PK, Taare Zameen Par

Since A and B have similar preferences, the system recommends Taare Zameen Par to user A.

Flow:-

Target User → Find similar users → check their likes
Recommend.

B. Item - Based Collaborative filtering.

It finds items that are similar based on users behavior.

For Ex, if many users who watched 3 Idiots also watched Taare Zameen Par, the system may recommend Taare Zameen Par to someone watching 3 Idiots.

Flow:

Users Item $\rightarrow$ Find Similar Items $\rightarrow$ Recommend similar items

Real world Example - Netflix

Consider Netflix.

suppose:

user Movies watched / Liked.

A Movie P, Movie Q, Movie R.

B Movie P, Movie Q, Movie R, Movie S.

C Movie X, Movie Y

Users A and user B have similar watching patterns.

Since user B also likes Movie S, Netflix can recommend Movie S to user A.

Thus, the recommendation is based on similar users behavior, rather than only analyzing the content of Movie S.

13. Differentiate between Linear Regression and Decision tree based on working, type of problem, advantages and limitations.

Basis	Linear Regression	Decision Tree
Working	Finds the best-fit straight line between input features and a continuous output	splits data into branches using conditions on features and makes predictions at leaf nodes.
Basic Idea	predicts output using a mathematical linear relationship.	predicts output by following a sequence of if-else conditions.
Type of problem	Mainly used for regression problems where output is numerical / continuous.	Can be used for both classification and regression.
Example	predicting house price from area, number of rooms etc.	predicting whether a student pass / fail based on attendance and marks.
Advantages	Simple, fast, easy to understand and interpret, works well when the relationship is approximately linear	Easy to understand and visualize, handles numerical and categorical features, captures non linear relationships, requires less data preprocessing.

Performs poorly when the relationship is non linear; sensitive to outliers; assumes a linear relationship between variables.

Can easily overfit; small changes in data can produce a different tree; large trees can become complex.

Easy to interpret using coefficients.

Very easy to interpret using the tree's branches and rules.

Usually lower risk with a simple model, but complex feature engineering can still cause issues.

Higher risk especially when the tree is allowed to grow very deep.

14. Explain the complete machine learning model development and evaluation process starting from dataset preparation to final model testing.

→ Machine learning model development is a step-by-step process of converting raw data into a model that can make accurate predictions on new data.

1. Dataset Collection and preparation:-

First, the required dataset is collected. The data is checked for missing values, duplicate records, incorrect values, and irrelevant data and cleaned properly.

2. Data preprocessing:-

The cleaned data is converted into a suitable form for the ML algorithm.

This may include:

- Handling missing values.
- Encoding categorical data into numbers.
- Scaling or normalizing numerical features.
- Removing unwanted noise.

3. Feature Selection

Important features are selected that have a useful relationship with the target variable.

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Irrelevant features are removed to reduce complexity and improve performance.

4. Train-Test split:

divides the dataset into:

- Training data - used to train the model.
- Testing data - used for final evaluation.

A validation set or k-fold cross-validation can also be used for model selection and tuning.

5. Model selection.

Choose a suitable algorithm according to the problem.

Ex:

- Classification → Decision Tree, KNN, SVM.
- Regression → Linear regression, Decision Tree regression.

6. Model Training:-

The training data is given to the selected algorithm. The model learns patterns and relationship between input features and the target output.

7. Model validation and Tuning.

Evaluate the model using validation data or cross-validation. Adjust hyperparameters to improve performance and reduce overfitting.

8. Final Model Testing

After selecting and tuning the model, test it on the unseen test dataset. This gives a more reliable estimate of how the model will perform on new data.

9. Performance Evaluation.

use suitable metrics:

- classification: Accuracy, Precision, Recall, F1 score
- Regression: MAE, MSE, RMSE, R^2

10. Deployment.

If the model gives satisfactory results, it can be deployed in a real world application to make predictions on new data.

15. Compare Overfitting, cross-validation, Instance-Based learning and feature reduction and explain how each helps in building an effective machine learning model.

→ These four concepts play different roles in building an effective Machine learning model. Overfitting is a problem, while cross-validation, Instance-based learning and feature reduction are techniques / concepts used in model development.

Concept	Meaning.	Main Purpose	Example
Overfitting	Model learns training data too closely, including noise	A problem to be avoided.	Model performs very well on training data but poorly on new data
Cross validation	Dividing data into multiple parts for training and validation.	Evaluate model reliability and reduce overfitting	K-Fold Cross Validation.
Instance based learning	Makes predictions using similarities with stored training examples.	Learn from existing instances	K-Nearest Neighbours (KNN)
Feature reduction	Reducing the number of input features.	Simplify the model and improve efficiency.	PCA, feature selection.

1. Overfitting

Overfitting occurs when a model becomes too closely fitted to the training data and also learns its noise and unnecessary details.

Ex: A student - performance model gives 98% accuracy on training data but only 70% on new student data.

2. Cross validation.

Cross validation is a technique used to evaluate how well a model performs on unseen data.

In k-Fold cross validation, the dataset is divided into k parts. The model is trained using k-1 parts and validated using the remaining part. This is repeated until every part has been used for validation.

3. Instance - Based Learning :-

Instance - Based learning stores training examples and makes predictions by comparing new data with similar stored examples.

A common example is k-Nearest Neighbors (KNN)

4. Feature Reduction.

Feature reduction means reducing the number of input variables while retaining the most useful information.

It can be done through:

- Feature selection - selecting important existing features
- Feature Extraction - creating a smaller set of new features, such as using PCA.

How they help build an effective model.

A good machine learning model should be accurate, simple, efficient and capable of working well on unseen data.

- Overfitting: Identifies a major problem that must be controlled.
- Cross valuation: checks whether the model generalizes well.
- Instance based - learning: provides a useful approach for prediction.

- Feature reduction - Makes the model simpler and can improve its efficiency and generalization.

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